

NEET Important Chapter – Magnetism and Matter

Magnetism and Matter for NEET Examination

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Magnetism and matter is one of the easiest and smallest chapters in Physics for the NEET exam. In magnetism and matter, we deal with magnetic dipoles and their properties in detail along with earth's magnetism. Students can easily score good marks from this chapter for the [NEET 2022](#) exam compared to other chapters.

The magnetism and matter chapter starts with magnetic dipoles and calculating the dipole moment of a magnetic dipole. Various terms related to magnetism like magnetic field lines, magnetising field, the intensity of magnetisation, etc. are discussed in this chapter. The formula to calculate the torque and potential energy of a bar magnet placed in an external magnetic field is also studied in this chapter.

In this chapter, we also learn what are the three elements of the earth's magnetic field. Materials are classified according to their magnetic properties like paramagnetic, diamagnetic and ferromagnetic. Curie's law and its equation are discussed in this chapter along with the hysteresis curve of ferromagnetic materials.

Now, let us move on to the important concepts and formulae related to NEET exams from magnetism and matter along with a few solved examples.

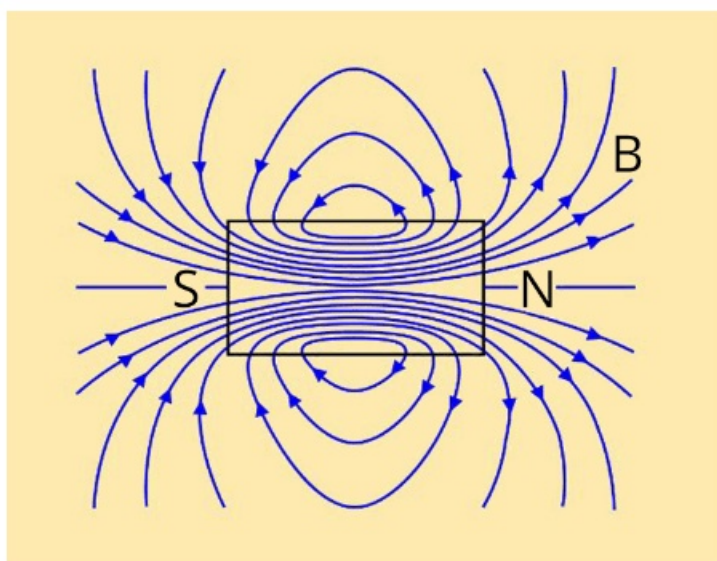
Important Topics of Magnetism and Matter

- Magnetic dipole moment of a magnet
- Terms related to magnetism
- Elements of Earth's Magnetic Field
- Magnetic Dipole Moment
- Magnetic Materials
- Magnetic field lines and their properties
- Gauss's Law of Magnetism
- Hysteresis curve
- Curie's Law

Important Concepts of Magnetism and Matter

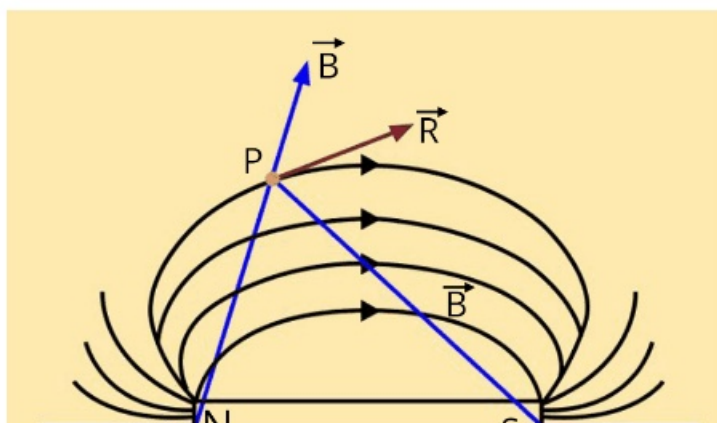
Name of the Concept	Key Points of the Concept
1. Magnetic dipole moment of a magnet	• Magnetic dipole moment of a magnet gives the strength of a magnet. • It is a vector quantity directed from south to north and its magnitude is given by Where m is the pole strength and l is the magnetic length. (image will be uploaded soon)

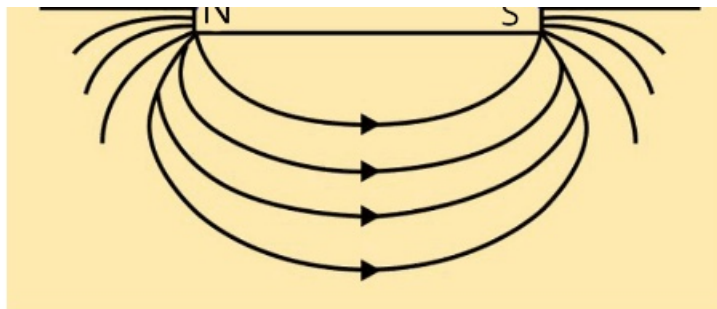
- The magnetic field produced by a bar magnet or magnetic substances can be represented using imaginary lines that represent the direction of the magnetic field at that point are called magnetic field lines.
- Magnetic field lines form continuous closed loops as shown in the diagram below.



2.

- The tangent to the magnetic field lines gives the direction of magnetic fields at that point.

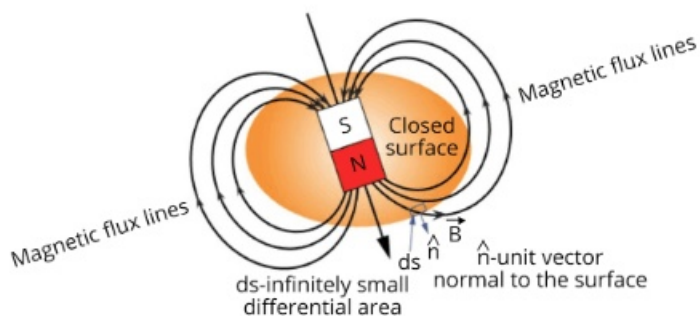




- Magnetic field lines never intersect each other.

3. Gauss's Law of Magnetism

- The net magnetic flux linked with the closed surface is zero.



- **Intensity of Magnetising Field (H):**
Intensity of magnetising field or magnetising field is the extent to which a magnetic field can magnetise a

4. Terms related to magnetism

substance and its SI unit is A/m

- **Intensity of Magnetisation(I):**

It is the degree to which a substance is magnetised when it is placed in a magnetic field given by the formula,

$$I = \frac{M}{V}$$

Where M is the induced dipole moment and V is the volume

- **Magnetic Susceptibility (χ_m)**

Magnetic susceptibility of a material is defined as the ratio of the intensity of magnetisation (I) to the magnetic intensity(H) applied to the substance.

$$\chi_m = \frac{I}{H}$$

- There are three elements used to completely determine the earth's magnetic field at a place. They are given below.

5. Elements of Earth's Magnetic Field

- Magnetic declination(θ): It is the angle between geographic meridian and magnetic meridian planes at a place.
- Angle of inclination or dip(ϕ): It is the angle that the intensity of the magnetic field of earth makes with horizontal line.
- The horizontal component of earth's magnetic field (B_H): The earth's magnetic field (B) at a place can be resolved into horizontal(B_H) and vertical (B_V) components.

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Where ϕ is the angle of inclination.

• Paramagnetic Materials

These materials experience a weak force of attraction in a magnetic field. They get feebly magnetised in the

6. Magnetic Materials

direction of the applied magnetic field.

(image will be uploaded soon)

Eg:- Al, Na, O₂ (at S.T.P)

- Diamagnetic materials

Diamagnetic materials experience weak force of repulsion in an external magnetic field. They also get weakly magnetised in the opposite direction of the applied magnetic field.

Eg: Cu, Bi, N₂ (at STP), Pd

(image will be uploaded soon)

- Ferromagnetic materials

These materials experience a strong force of attraction in a magnetic field. They get strongly magnetised in the direction of the applied magnetic field.

Eg: Fe, Co, Ni,

- According to Curie's law, magnetic susceptibility of a paramagnetic substance is inversely proportional to its absolute temperature.

7. Curie's Law

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Where C is the Curie's constant and T is the absolute temperature.

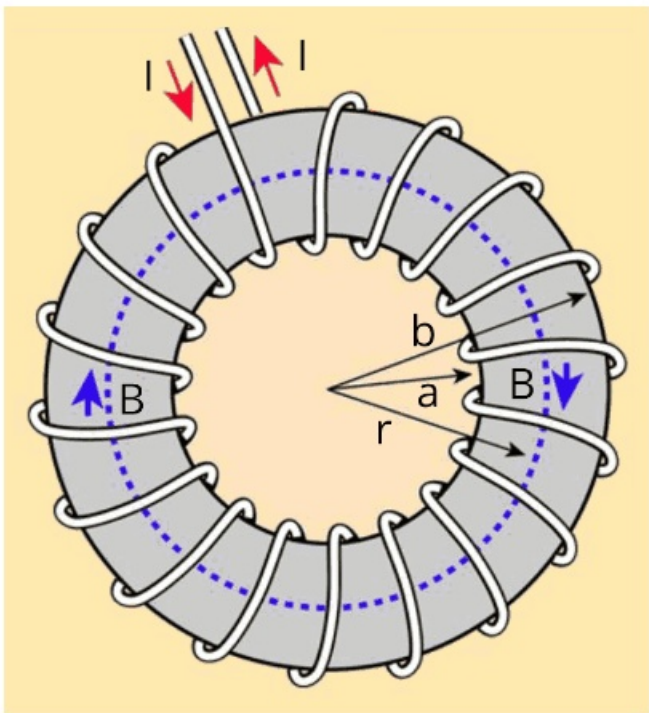
- The temperature above which a ferromagnetic material becomes paramagnetic material is called Curie temperature.

- It is a B-H curve of a ferromagnetic material by noting down the variation in magnetic field density (B) with respect to changing magnetic field (H).

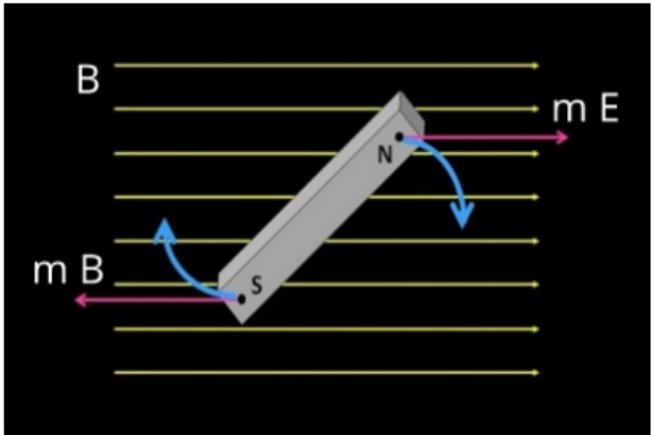
8. Hysteresis Curve	(image will be uploaded soon) • The lagging of magnetic flux density behind the magnetising field is called hysteresis and the B-H curve is called the hysteresis curve. • The remainder value of magnetic flux density (B_r), even after magnetising field (H) having it reduced to zero, is called retentivity. • The value of magnetising in the opposite direction to reduce the magnetisation of the material to zero is called coercivity • The area of the B-H curve gives the energy loss per one cycle of magnetisation and demagnetisation.
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List of Important Formulae

Sl. No	Name of the Concept	Formula
1.	Lorentz force formula	The force acting on a charged particle (q) in a combined electric field (E) and magnetic field (B)

2.	Pitch and radius of a helical path of a charged particle (q) moving at an angle θ to the magnetic field (B) with a velocity (v)	_____ _____
3.	Magnetic field inside a toroid	 The diagram illustrates a toroid, which is a ring-shaped magnetic core with a grey cross-section. It is wound with a series of white coils. Red arrows labeled 'I' indicate the direction of current flow into and out of the coils. A dashed blue line with arrows represents the magnetic field (B) circulating within the central cavity of the toroid. A radial line from the center to the inner edge is labeled 'a', the radial line to the outer edge is labeled 'b', and the mean radius is labeled 'r'.
4.	The magnetic moment of a current-carrying coil having an area of cross-section (A) and N number of turns	
5.	The torque acting on a current carrying coil placed in a magnetic field B	Where θ is the angle between area vector and magnetic field and N is the number of turns

	magnetic field B	or turns
6.	The torque acting on a magnetic dipole placed in a magnetic field B	Where θ is the angle between magnetic dipole moment M and magnetic field B .

7.	The potential energy of a magnetic dipole in an external magnetic field B .	Where θ is the angle between magnetic dipole moment M and magnetic field B. 
8.	Relation between relative permeability (μ_r) and magnetic susceptibility (χ_m)	

Solved Examples

1. A magnetic needle lying parallel to a magnetic field requires W units of work to turn it through 60° . The torque required to maintain the needle in this position in terms of W will be

Sol:

Ex 1.2

The work done to rotate the magnetic needle when it is parallel to the magnetic field to an angle of 60° can be calculated using the formula given below.

$$\text{Work done} = \dots (1)$$

The torque required to maintain the needle can be calculated using the formula given by

$$\text{Torque} = \dots (2)$$

Divide equation (2) by equation (1) to get the torque in terms of W.

$$\frac{\text{Torque}}{\text{Work done}} = \frac{\dots}{\dots}$$

Key Point: To maintain the needle placed in a magnetic field at a particular position, an equal and opposite torque experienced by the needle has to be applied to it.

2. If the angle of dip at two places are 30° and 45° respectively, then the ratio of the horizontal components of earth's magnetic field at two places will be

Sol:

Given:

The angle of dip of two places are $\theta_1 = 30^\circ$ and $\theta_2 = 45^\circ$

The horizontal component of earth's magnetic field where $\theta_1 = 30^\circ$ is given by

$$\frac{B \sin \theta}{B \cos \theta} = \tan \theta \quad \dots(1)$$

Similarly, the horizontal component of earth's magnetic field where $\theta_1 = 45^\circ$ is given by

$$B \cos \theta \quad \dots(2)$$

Divide equation (1) by equation (2) to obtain the ratio of horizontal components of earth's magnetic field.

$$\frac{B \sin \theta}{B \cos \theta} = \tan \theta$$

Key Point: The horizontal component of earth's magnetic field and dip angle is related by the formula.

Previous Year Questions from NEET Paper

1. A magnetic needle suspended parallel to a magnetic field requires $\sqrt{3}$ J of work to turn it through 60° . The torque needed to maintain the needle in this position will be (NEET 2012)

a. $2\sqrt{3}$ J

b. 3 J

c. $\sqrt{3}$ J

d. $3/2$ J

Sol:

The work done to rotate the magnetic needle when it is parallel to the magnetic field to an angle of 60° can be calculated using the formula given below.

$$- \quad \text{---} \dots (1)$$

The torque required to maintain the needle can be calculated using the formula given by

$$- \quad \text{---} \dots (2)$$

Divide equation (2) by equation (1) to calculate the torque.

$$\frac{-}{-} = \frac{\text{---}}{\text{---}}$$

Therefore, correct option is option (b)

Trick: To maintain the needle placed in a magnetic field at a particular position, an equal and opposite torque experienced by the needle has to be applied to it.

2. An iron rod of susceptibility 599 is subjected to a magnetising field of 1200 Am^{-1} . The permeability of the material of the rod is (NEET 2020)

a. $2.4\pi \times 10^{-5} \text{ TmA}^{-1}$

b. $2.4\pi \times 10^{-7} \text{ TmA}^{-1}$

c. $2.4\pi \times 10^{-4} \text{ TmA}^{-1}$

d. $8 \times 10^{-5} \text{ TmA}^{-1}$

Sol:

The relative magnetic permeability of the material can be calculated using the formula given by

The permeability of the material is calculated by the following formula.

Trick: The relative magnetic permeability is the ratio of magnetic permeability of the material to the magnetic permeability of free space or vacuum.

Practice Questions

1. When a piece of a ferromagnetic substance is put in a uniform magnetic field, the flux density inside it is four times the flux density away from the piece. The magnetic permeability of the material is (**Ans: 4**)

2. A short bar magnet of magnetic moment 0.4 JT^{-1} is placed in a uniform magnetic field of 0.16 T . The magnet is in stable equilibrium when the potential energy is (**Ans: 0.064J**)

Conclusion

Conclusion

In this article, we have provided important information regarding the chapter Magnetism and Matter, such as important concepts, formulae, etc. Students should work on more solved examples along with previous year question papers for scoring good grades in the NEET exams.

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